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1.

i. $\dfrac{d^2y}{dx^2} - 3x\sin y' = x^3 + 4,$

ii. $\dfrac{\partial^3 z}{\partial x^2 \partial t} + z^2 \dfrac{\partial^2 z}{\partial x \partial t} + \dfrac{\partial^2 z}{\partial t^2} = \cos(x^2),$

iii. $(x^2 + \sin(y))z_x + \sqrt{x} z_y = 4x + 6y^3,$

iv. $zz_x - (xz^2 + y)z_y = -z^2 x + y,$

v. $x^2 z_x + (x^2 + 3y)z_y + 4z = \cot(zy)$

Yukarıdaki denklemlerin tipini belirleyiniz. (Adi, kısmi, lineer, yarı lineeer, hemen hemen lineer ve lineer olmayan denklemler şeklinde sadece sınıflandırınız)

2. $xyz = f(x + y + z)$ yüzey aileleri tarafından temsil edilen en küçük mertebeden kısmi denklemi elde ediniz

3. $x^2 u_{xx} + 2xyu_{xy} + y^2 u_{yy} = 4x^2$ kanonik forma indirgeyip çözünüz?

4. $(y + 2ux)u_x - (x + 2uy)u_y = \frac{1}{2}(x^2 - y^2)$ denkleminin genel çözümünü bulunuz.

Cevab2.

$$yz + xyz_x = (z_x + 1)f'$$

$$xz + xyz_y = (z_y + 1)f'$$

$$\frac{yz + xyz_x}{z_x + 1} = \frac{xz + xyz_y}{z_y + 1}$$

$$yz + xyz_x + yzz_y + xyz_x z_y = xz + yxz_y + xzz_x + xyz_x z_y$$

$$\Rightarrow (1 + q)yz + (1 + q)xyp = (1 + p)xz + (1 + p)xyq$$

y(z-x)q+x(y-z)p=z(x-y)

3. $x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy} = 4x^2$ kanonik forma indirgeyip çözünüz.

$A = x^2 \quad B = 2xy \quad C = y^2$

$\Delta = B^2 - 4AC = 4x^2y^2 - 4x^2y^2 = 0$ parabolik

$$\frac{dy}{dx} = \frac{B}{2A} = \frac{2xy}{2\cdot x^2} \qquad \frac{dy}{dx} = \frac{y}{x} \qquad \frac{dy}{y} - \frac{dx}{x} = 0$$

$$\frac{y}{x} = c_1 = \xi$$

$$y = c_2 = \eta$$

$$\left.\begin{aligned} \xi &= \frac{y}{x} \\ \eta &= y \end{aligned}\right\} \frac{\partial \xi \eta}{\partial xy} = \begin{vmatrix} -\frac{y}{x^2} & \frac{1}{x} \\ 0 & 1 \end{vmatrix} = -\frac{y}{x^2} \neq 0$$

$$u_x = u_\xi \xi_x + \cancel{u_\eta \eta_x}^{\,0} = -\frac{y}{x^2} u_\xi$$

$$u_{xx} = \frac{2y}{x^3} u_\xi - \frac{y}{x^2}\left(u_{\xi\xi}\xi_x + \cancel{u_{\xi\eta}\eta_x}^{\,0}\right)$$

$$= \frac{2y}{x^3} + \frac{y^2}{x^4} u_{\xi\xi}$$

$$u_{xy} = -\frac{1}{x^2} u_\xi - \frac{y}{x^2}\left(u_{\xi\xi}\underbrace{\xi_y}_{\frac{1}{x}} + u_{\xi\eta}\eta_y\right)$$

$$= -\frac{1}{x^2} u_\xi - \frac{y}{x^3} u_{\xi\xi} - \frac{y}{x^2} u_{\xi\eta}$$

$$u_y = u_\xi \xi_y + u_\eta \eta_y = \frac{1}{x} u_\xi + u_\eta$$

$$u_{yy} = \frac{1}{x}\left(u_{\xi\xi}\xi_y + u_{\xi\eta}\eta_y\right) + u_{\eta\eta}\eta_y + u_{\eta\xi}\xi_y$$

$$= \frac{1}{x^2} u_{\xi\xi} + \frac{1}{x} u_{\xi\eta} + u_{\eta\eta} + u_{\eta\xi}\frac{1}{x}$$

$$= \frac{1}{x^2} u_{\xi\xi} + \frac{2}{x} u_{\xi\eta} + u_{\eta\eta}$$

$$x^2\left(\frac{2y}{x^3}u_\xi+\frac{y^2}{x^4}u_{\xi\xi}\right)+2xy\left(-\frac{1}{x^2}u_\xi-\frac{y}{x^3}u_{\xi\xi}-\frac{y}{x^2}u_{\xi\eta}\right)$$

$$+y^2\left(\frac{1}{x^2}u_{\xi\xi}+\frac{2}{x}u_{\xi\eta}+u_{\eta\eta}\right)=4x^2$$

$$\cancelto{0}{\frac{2y}{x}u_\xi}+\cancelto{0}{\frac{y^2}{x^2}u_{\xi\xi}}-\cancelto{0}{\frac{2y}{x}u_\xi}-\cancelto{0}{\frac{2y^2}{x^2}u_{\xi\xi}}-\cancelto{0}{\frac{2y^2}{x}u_{\xi\eta}}$$

$$+\cancelto{0}{\frac{y^2}{x^2}u_{\xi\xi}}+\cancelto{0}{\frac{2y^2}{x}u_{\xi\eta}}+y^2u_{\eta\eta}=4x^2$$

$$\eta=y$$

$$\xi=\frac{y}{x}\qquad x=\frac{\eta}{\xi}$$

$$y^2u_{\eta\eta}=4x^2$$

$$\cancel{\eta^2}u_{\eta\eta}=4\frac{\cancel{\eta^2}}{\xi^2}$$

$$u_{\eta\eta}=\frac{4}{\xi^2}$$

$$u_\eta=\frac{4\eta}{\xi^2}+f(\xi)$$

$$u=\frac{2\eta^2}{\xi^2}+\eta f_1(\xi)+f_2(\xi)$$

$$u=2x^2+yf_1\left(\frac{y}{x}\right)+f_2\left(\frac{y}{x}\right)$$

$$(y+2ux)u_x-(x+2uy)u_y=\frac{1}{2}(x^2-y^2)$$

$$\underbrace{\frac{dx}{y+2ux}=\frac{dy}{-(x+2uy)}}=\frac{du}{\frac{1}{2}(x^2-y^2)}$$

$$\Downarrow$$

$$\frac{x\,dx+y\,dy}{2u(x^2-y^2)}=\frac{2du}{x^2-y^2}\Rightarrow \varphi(x,y,u)=x^2+y^2-4u^2=c_1$$

$$\frac{y\,dx+x\,dy}{y^2-x^2}=\frac{2du}{x^2-y^2}\Rightarrow \psi(x,y,u)=xy+2u=c_2$$

<u>Sol.</u>: $F(x^2+y^2-4u^2,\ xy+2u)=0$

or

$$x^2+y^2-4u^2=f(xy+2u)$$